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United States Patent	12418142
Kind Code	B2
Date of Patent	September 16, 2025
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Connector

Abstract

Technology of the present invention improves the shielding performance of an outer conductor. A connector **10** includes an inner conductor **20** and an outer conductor **40** that surrounds the inner conductor **20**. The outer conductor **40** includes a first outer conductor **41** that can be electrically connected to a shield layer **83** of an electric wire **80**, and a second outer conductor **42** that can be electrically connected to the first outer conductor **41** and can be electrically connected to a partner outer conductor **93** of a partner connector **90**. The first outer conductor **41** is a member formed with a tubular shape by casting or cutting, and includes a first lock portion **47**. The second outer conductor **42** is a plate-shaped member and includes a second lock portion **51** that can be locked to the first lock portion **41.**

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Appl. No.: **17/949916**

Filed: **September 21, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230113071 A1	Apr. 13, 2023

Foreign Application Priority Data

JP	2021-157711	Sep. 28, 2021

Publication Classification

Int. Cl.: **H01R13/627** (20060101); **H01R43/042** (20060101)

U.S. Cl.:

CPC **H01R13/6273** (20130101); **H01R43/0421** (20130101);

Field of Classification Search

CPC: H01R (9/0524); H01R (9/0527); H01R (9/0518); H01R (13/6273); H01R (43/0421); H01R (2103/00)

USPC: 439/582; 439/535

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority from Japanese Patent Application No. 2021-157711, filed on Sep. 28, 2021, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a connector.

BACKGROUND

(3) Japanese Patent Laid-open Publication No. 2000-331754 discloses a connector that includes an

outer conductor. The outer conductor includes a metal body case and a metal leading-end tubular case that are connected to each other. The metal body case is connected to an antenna wire, and the metal leading-end tubular case is connected to a partner socket. The metal body case and the metal leading-end tubular case are each plate-shaped. A connector that includes an outer conductor is also disclosed in Japanese Patent Laid-open Publication Nos. S63-021769, H05-129053 and 2011-134720.

SUMMARY

(4) In the connector described above, in the case where the outer conductor is formed by bending a metal plate, there is a concern of the formation of a gap and that shielding performance cannot be ensured.

(5) In view of this, an object of the present disclosure is to provide a technique capable of improving the shielding performance of an outer conductor.

(6) A connector according to the present disclosure includes: an inner conductor; and an outer conductor configured to surround the inner conductor, wherein the outer conductor includes: a first outer conductor configured to be electrically connected to a shield layer of an electric wire, and a second outer conductor configured to be electrically connected to the first outer conductor and electrically connected to a partner outer conductor of a partner connector, the first outer conductor is a member formed with a tubular shape by casting or cutting, and includes a first lock portion, and the second outer conductor is a plate-shaped member and includes a second lock portion configured to be locked to the first lock portion.

(7) According to the present disclosure, the shielding performance of the outer conductor can be improved.

(8) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view of a connector according to a first embodiment.

(2) FIG. **2** is a perspective view of an inner conductor, an outer conductor, and a dielectric in an assembled state.

(3) FIG. **3** is a perspective view showing a state before a second outer conductor has been fitted to a first outer conductor.

(4) FIG. **4** is a perspective view of an inner conductor connected to an electric wire.

(5) FIG. **5** is a side cross-sectional view of the connector and a partner connector.

(6) FIG. **6** is an enlarged view of a region Z shown in FIG. **5**.

(7) FIG. **7** is a cross-sectional view taken along line A-A in FIG. **5**.

(8) FIG. **8** is a cross-sectional view taken along line B-B in FIG. **5**.

(9) FIG. **9** is a cross-sectional view of a region including the inner conductor, the outer conductor, and the dielectric, at a cross-section taken along line C-C in FIG. **5**.

(10) FIG. **10** is a cross-sectional view of a region including a first inner conductor and a first dielectric, at a cross-section taken along line D-D in FIG. **6**.

(11) FIG. **11** is a perspective view showing a state before another type of second outer conductor has been fitted to the first outer conductor.

DETAILED DESCRIPTION

(12) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings,

and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.

Description of Embodiments of Present Disclosure

(13) First, embodiments of the present disclosure will be listed and described.

(14) (1) A connector according to the present disclosure includes: an inner conductor; and an outer conductor configured to surround the inner conductor, wherein the outer conductor includes: a first outer conductor configured to be electrically connected to a shield layer of an electric wire, and a second outer conductor configured to be electrically connected to the first outer conductor and electrically connected to a partner outer conductor of a partner connector, the first outer conductor is a member formed with a tubular shape by casting or cutting, and includes a first lock portion, and the second outer conductor is a plate-shaped member and includes a second lock portion configured to be locked to the first lock portion.

(15) According to this connector, the first outer conductor is a member formed with a tubular shape by casting or cutting, and therefore the first outer conductor can be formed so as to suppress the formation of a gap, thus making it possible to improve the shielding performance of the first outer conductor.

(16) Here, if the second outer conductor is also a member formed with a tubular shape by casting or cutting, both the first outer conductor and the second outer conductor do not easily deform, and thus can conceivably be coupled to each other by press fitting. However, in the case of coupling by press fitting, the dimensional tolerance between the first outer conductor and the second outer conductor needs to be reduced in order to ensure electrical connection reliability, which may increase difficulty in manufacturing the outer conductor. In view of this, according to the above connector, the second outer conductor is a plate-shaped member and includes the second lock portion that can undergo bending deformation. Therefore, if the second lock portion undergoes bending deformation in order to become locked to the first lock portion of the first outer conductor, the dimensional tolerance between the first outer conductor and the second outer conductor can be large, the outer conductor can be manufactured easily, and it is possible to realize coupling with high electrical connection reliability. Also, due to being plate-shaped, the second outer conductor can be manufactured at low cost.

(17) Also, in the case of manufacturing a plurality of types of connectors in correspondence with various types of partner connectors, by preparing a plurality of types of second outer conductors in correspondence with the types of partner outer conductors of the partner connectors, it possible to manufacture a plurality of types of connectors that can be fitted to various types of partner outer conductors while also including the same first outer conductor.

(18) (2) It is preferable that one of a plurality of types of the second outer conductor is selectively coupled to the first outer conductor, and the plurality of types of second outer conductors include partner connection portions having different shapes from each other.

(19) According to this configuration, it is possible to manufacture a plurality of types of connectors that can be fitted to various types of partner outer conductors while also including the same first outer conductor.

Detailed Description of Embodiments of Present Disclosure

(20) Specific examples of the present disclosure will be described below with reference to the drawings. It should be noted that the present invention is not limited to these examples, but rather is indicated by the scope of claims, and is intended to include all modifications within a meaning and scope equivalent to the scope of claims.

First Embodiment

(21) FIG. **1** shows a connector **10** of a first embodiment. In the following description, the up-down direction shown in FIG. **5** is defined as-is as the up-down direction of the connector **10**. Also, the left side shown in FIG. **5** is the front side of the connector **10**, and the right side is the rear side of the connector **10**. Moreover, the left-right direction in a front view of the connector **10** is the left-

right direction of the connector **10**.
(22) Overview of Connector **10**
(23) As shown in FIG. **1**, the connector **10** is L-shaped. As shown in FIG. **5**, a partner connector **90** is fitted to one end side of the connector **10**, and an electric wire **80** is electrically connected to the other end side of the connector **10**. The electric wire **80** is a shielded electric wire, and is configured as a coaxial cable in the present embodiment. The electric wire **80** includes an inner conductor **81**, an insulator **82**, a shield layer **83**, and a sheath **84**. The insulator **82** surrounds the inner conductor **81**. The shield layer **83** surrounds the insulator **82**. The sheath **84** surrounds the shield layer **83**. The partner connector **90** includes a partner housing **91**, a partner inner conductor **92**, and a partner outer conductor **93**.
(24) As shown in FIG. **5**, the connector **10** includes a housing **11**, an inner conductor **20**, an outer conductor **40**, a dielectric **60**, a sleeve **70**, a first seal member **71**, a second seal member **72**, and retaining members **73**.
(25) Configuration of Housing **11**
(26) The housing **11** is electrically insulating and is made of a synthetic resin. The housing **11** is L-shaped as shown in FIG. **1**. As shown in FIGS. **1** and **5**, the housing **11** includes a housing body **12**, a fitting hole **13**, a fitting groove **14**, an inner hood portion **15**, an outer hood portion **16**, a lock arm **17**, and first retaining and locking portions **18**.
(27) As shown in FIG. **5**, the housing body **12** is shaped as tube (more specifically, a square tube) that extends in the up-down direction. The lower end of the housing body **12** is open downward, and the upper end is closed.
(28) As shown in FIG. **5**, the fitting hole **13** is formed so as to extend from the inner peripheral surface of the housing body **12** to the outside. In other words, the fitting hole **13** extends in the front-rear direction through a wall portion (the front wall in the present embodiment) of the housing body **12**. The fitting hole **13** is open on the front side of the housing **11**. The fitting hole **13** is provided on the upper end side of the center of the housing body **12** in the up-down direction. The outer conductor **40** is fitted into the fitting hole **13**.
(29) As shown in FIG. **5**, the fitting groove **14** extends along the front-rear direction in the inner peripheral surface of the fitting hole **13**. The fitting groove **14** is open on the front and rear sides.
(30) As shown in FIG. **5**, the inner hood portion **15** is shaped as a tube that projects forward from a portion of the housing body **12** that surrounds the fitting hole **13**. The inner hood portion **15** is shaped as a tube (more specifically, a cylinder) that extends in the front-rear direction. The inner space of the inner hood portion **15** is in communication with the fitting hole **13** and is open on the front side of the housing **11**.
(31) As shown in FIG. **5**, the outer hood portion **16** surrounds the outer peripheral surface of the inner hood portion **15**. The outer hood portion **16** is shaped as a tube that extends in the front-rear direction. The inner space of the outer hood portion **16** is open on the front side of the housing **11**. The front end of the outer hood portion **16** is located forward of the front end of the inner hood portion **15**.
(32) As shown in FIG. **5**, the lock arm **17** is located inside the outer hood portion **16**. The lock arm **17** is shaped so as to extend in the front-rear direction. The front end side of the lock arm **17** is supported so as to be swingable in the up-down direction. The lock arm **17** is supported by the outer hood portion **16** as shown in FIGS. **7** and **8**. The lock arm **17** is locked to a partner locking portion **94** of the partner housing **91** of the partner connector **90** (see FIG. **5**).
(33) As shown in FIG. **1**, the first retaining and locking portions **18** are shaped so as to project from outer peripheral faces (left and right side faces in the present embodiment) of the housing body **12**. The first retaining and locking portions **18** are provided in a lower end portion of the housing body **12**. The retaining members **73** engage with the first retaining and locking portions **18**.
(34) Overview of Inner Conductor **20**, Outer Conductor **40**, and Dielectric **60**
(35) As shown in FIG. **4**, the inner conductor **20** is L-shaped. The inner conductor **20** includes a

first inner conductor **21** and a second inner conductor **22**. The first inner conductor **21** and the second inner conductor **22** are each made of a metal, and are formed by bending a metal plate. The first inner conductor **21** and the second inner conductor **22** are electrically connected to each other.
(36) As shown in FIG. **5**, the outer conductor **40** is L-shaped and surrounds the inner conductor **20**. The outer conductor **40** includes a first outer conductor **41** and a second outer conductor **42**. The first outer conductor **41** and the second outer conductor **42** are each made of a metal. The first outer conductor **41** and the second outer conductor **42** are electrically connected to each other.
(37) The dielectric **60** has an insulating characteristic and is made of a synthetic resin. As shown in FIG. **5**, the dielectric **60** is arranged between the inner conductor **20** and the outer conductor **40**. The dielectric **60** includes a first dielectric **61** and a second dielectric **62**.
(38) Overview of First Inner Conductor **21**
(39) The first inner conductor **21** is a plate-shaped member, and is formed by bending a metal plate. As shown in FIGS. **4** to **6**, the first inner conductor **21** is shaped so as to extend in the up-down direction. The first inner conductor **21** includes a first inner conductor body **23**, a locking portion **24**, a stabilizer **25**, and a barrel portion **26**.
(40) As shown in FIGS. **4**, **6** and **10**, the first inner conductor body **23** includes a first bottom plate portion **28**, a second bottom plate portion **29**, a pair of side plate portions **30**, a pair of first connection portions **31**, and a pair of invitation portions **32**. The first bottom plate portion **28** and the second bottom plate portion **29** are spaced apart from each other in the up-down direction. The thickness directions of the first bottom plate portion **28** and the second bottom plate portion **29** conform to the front-rear direction. The pair of side plate portions **30** are respectively continuous with the left and right sides of the first bottom plate portion **28** and the second bottom plate portion **29** and project forward from the same. The pair of side plate portions **30** are spaced apart from each other in the left-right direction. The pair of first connection portions **31** extend forward from the front ends of the pair of side plate portions **30**. The pair of first connection portions **31** extend from portions, with respect to the up-down direction, of the front ends of the pair of side plate portions **30**. The shortest gap between the pair of first connection portions **31** is smaller than the gap between the pair of side plate portions **30**. The pair of guide portions **32** extend forward from the front ends of the pair of first connection portions **31**. The distance between the pair of guide portions **32** increases as they extend forward.
(41) As shown in FIG. **6**, the locking portion **24** extends in the up-down direction. The locking portion **24** is shaped as a double-supported beam whose upper and lower end portions are supported by the first inner conductor body **23**. The lower end portion of the locking portion **24** is supported by the upper end portion of the first bottom plate portion **28**, and the upper end portion of the locking portion **24** is supported by the lower end portion of the second bottom plate portion **29**. The locking portion **24** is plate-shaped and can undergo bending deformation in the front-rear direction. The thickness direction of the locking portion **24** conforms to the front-rear direction. The locking portion **24** bulges rearward from the first inner conductor body **23**. The locking portion **24** is curved. The locking portion **24** has a protruding face **24**A that bulges from the first inner conductor body **23** and a receding face **24**B formed on the reverse side of the protruding face **24**A. In other words, the protruding face **24**A is formed on the rear face of the locking portion **24**, and the receding face **24**B is formed on the front face of the locking portion **24**.
(42) As shown in FIGS. **4** and **10**, the stabilizer **25** is provided at the front end of one side plate portion **30** (in the present embodiment, the right side plate portion **30**) out of the two of side plate portions **30**. The stabilizer **25** is spaced apart from the pair of first connection portions **31** in the up-down direction. More specifically, the stabilizer **25** is arranged below the first connection portions **31**. The stabilizer **25** is bent so as to project outward in the left-right direction.
(43) As shown in FIG. **4**, the barrel portion **26** is crimped to the inner conductor **81** of the electric wire **80** and is electrically connected to the inner conductor **81**.
(44) Configuration of First Outer Conductor **41**

(45) The first outer conductor **41** is a member formed with a tubular shape by casting or cutting. "Formed with a tubular shape by casting or cutting" means perform the step of formation with a tubular shape by performing casting or cutting, and does not mean formation with a tubular shape by bending a metal plate that has been cut. Note that casting also includes die casting. As shown in FIG. **5**, the first outer conductor **41** surrounds the first inner conductor **21**. As shown in FIGS. **3** and **5**, the first outer conductor **41** includes an accommodation portion **43**, a tubular portion **44**, a conductor-side fitting hole **45**, a conductor-side fitting groove **46**, first lock portions **47**, through holes **48**, and sleeve positioning portions **49**.
(46) As shown in FIGS. **3** and **5**, the accommodation portion **43** is formed by forming an opening in one end side of the first outer conductor **41**. The opening direction of the accommodation portion **43** is the forward direction. The tubular portion **44** extends in the up-down direction. The tubular portion **44** is formed by forming an opening on the other end side of the first outer conductor **41**. The opening direction of the tubular portion **44** is the downward direction. In other words, the opening direction of the accommodation portion **43** intersects (in the present embodiment, is orthogonal to) the opening direction of the tubular portion **44**. The inner space of the first outer conductor **41** is shaped such that the space extending rearward from the opening of the accommodation portion **43** and the space extending upward from the opening of the tubular portion **44** are orthogonal to each other.
(47) The conductor-side fitting hole **45** is formed inside the accommodation portion **43** as shown in FIGS. **3** and **5**. The conductor-side fitting hole **45** passes through a peripheral wall of the tubular portion **44** and is in communication with the inner space of the tubular portion **44**. The conductor-side fitting hole **45** is open on the front side of the first outer conductor **41**. The conductor-side fitting hole **45** is arranged on the upper end side of the center of the first outer conductor **41** in the up-down direction. The conductor-side fitting groove **46** extends along the front-rear direction in the inner peripheral surface of the conductor-side fitting hole **45**. The conductor-side fitting groove **46** is open on the front and rear sides.
(48) As shown in FIGS. **3** and **9**, the first lock portions **47** are provided inward of the conductor-side fitting hole **45** in the accommodation portion **43**. A pair of first lock portions **47** are provided on the left and right sides. The first lock portions **47** are shaped so as to project inward from inner peripheral faces of the accommodation portion **43**. The front faces of the first lock portions **47** are inclined rearward while extending toward the center of the accommodation portion **43** in a view from the front. The rear faces of the first lock portions **47** extend along the up-down direction and the left-right direction.
(49) As shown in FIGS. **3** and **9**, the through holes **48** are formed at positions corresponding to the pair of first lock portions **47**. In the present embodiment, the "positions corresponding to the pair of first lock portions" are rearward of the first lock portions **47** (i.e., deeper than the first lock portions **47** inside the accommodation portion **43**). The through holes **48** connect the inner space of the accommodation portion **43** to the space outside. The through holes **48** are die-cut holes formed when the first outer conductor **41** is manufactured.
(50) As shown in FIG. **2**, the sleeve positioning portions **49** are shaped so as to project from outer peripheral faces of the tubular portion **44**. The sleeve positioning portions **49** respectively project from the left and right sides of the tubular portion **44**. As shown in FIG. **8**, the sleeve **70** is arranged below the sleeve positioning portions **49**. The sleeve positioning portions **49** restrict upward movement of the sleeve **70**.
(51) Configuration of First Dielectric **61**
(52) As shown in FIGS. **5**, **6** and **10**, the first dielectric **61** is arranged between the first inner conductor **21** and the first outer conductor **41**. The first dielectric **61** includes a cavity **63**, a locking hole **64**, an entry hole **65**, a guide groove **66**, and a stabilizer fitting groove **67**.
(53) The cavity **63** extends along the up-down direction as shown in FIG. **5**. The cavity **63** is open on the lower side of the first dielectric **61**.

(54) As shown in FIG. **6**, the locking hole **64** is formed in an inner peripheral face of the cavity **63** (more specifically, the rearward inner peripheral face). The locking portion **24** of the first inner conductor **21** enters the locking hole **64**.
(55) As shown in FIG. **6**, the entry hole **65** is formed in an inner peripheral face of the cavity **63** (more specifically, the forward inner peripheral face). The entry hole **65** is formed at a position opposing the locking hole **64** in the front-rear direction. The second inner conductor **22** enters the entry hole **65** and is arranged therein. The locking hole **64** and the entry hole **65** are arranged coaxially with each other via the cavity **63**.
(56) As shown in FIG. **6**, the guide groove **66** is formed in an inner peripheral face of the cavity **63**. The guide groove **66** extends along the up-down direction and is in communication with the locking hole **64**.
(57) As shown in FIG. **10**, the stabilizer fitting groove **67** is formed in an inner peripheral face of the cavity **63**. The stabilizer fitting groove **67** extends along the up-down direction. The stabilizer fitting groove **67** is formed at a position corresponding to the stabilizer **25** of the first inner conductor **21**. The stabilizer **25** of the first inner conductor **21** enters the stabilizer fitting groove **67**.
(58) Configuration of Second Inner Conductor **22**
(59) The second inner conductor **22** is a plate-shaped member, and is formed by bending a metal plate. The second inner conductor **22** extends in the front-rear direction as shown in FIGS. **4** and **6**. The second inner conductor **22** includes a second inner conductor body **34**, an inner-conductor-side partner connection portion **35**, a second connection portion **36**, inner-conductor-side protruding portions **37**, and retaining projections **38**.
(60) As shown in FIGS. **4** and **6**, the second inner conductor body **34** is shaped as a tube (more specifically, a cylinder) that extends in the front-rear direction.
(61) As shown in FIGS. **4** and **6**, the inner-conductor-side partner connection portion **35** is located forward of the second inner conductor body **34**. The inner-conductor-side partner connection portion **35** is electrically connected to the partner inner conductor **92** (see FIG. **5**) of the partner connector **90**.
(62) As shown in FIGS. **4** and **6**, the second connection portion **36** is located rearward of the second inner conductor body **34**. The second connection portion **36** is configured as a tab. The second connection portion **36** projects rearward from the rear end of the second dielectric **62**. The second connection portion **36** is electrically connected to the first connection portion **31** of the first inner conductor **21**.
(63) As shown in FIG. **6**, the inner-conductor-side protruding portions **37** are provided on the outer peripheral surface of the second inner conductor body **34** and project upward from the outer peripheral surface. When the second inner conductor **22** is inserted into the second dielectric **62** and the inner-conductor-side protruding portion **37** comes into contact with the rear face of the second dielectric **62**, the second inner conductor **22** is restricted from moving forward of the second dielectric **62**.
(64) As shown in FIG. **9**, the retaining projections **38** are respectively provided on the left and right sides of the second inner conductor body **34** and project outward in the left-right direction. The retaining projections **38** prevent the second inner conductor **22** from coming out rearward when normally inserted into the second dielectric **62**.
(65) Configuration of Second Outer Conductor **42**
(66) The second outer conductor **42** is a plate-shaped member, and is formed by bending a metal plate. The second outer conductor **42** surrounds the second inner conductor **22** as shown in FIG. **6**. The second outer conductor **42** is shaped as a tube (more specifically, a cylinder) that extends in the front-rear direction. The second outer conductor **42** is open on the front and rear sides. As shown in FIG. **3**, the second outer conductor **42** includes an outer conductor body **50**, second lock portions **51**, protruding portions **52**, a front stop portion **53**, and partner connection portions **54**.
(67) As shown in FIG. **3**, the outer conductor body **50** is shaped as a tube (more specifically, a

cylinder).
(68) As shown in FIGS. **3** and **9**, the second lock portions **51** project rearward from the outer conductor body **50**. The second lock portions **51** are supported in a cantilevered manner by the outer conductor body **50**. The second lock portions **51** are respectively provided on the left and right sides of the outer conductor body **50**. The second lock portions **51** are plate-shaped and can undergo bending deformation toward the center of the second outer conductor **42** (i.e., radially inward) in a view from the front. The second lock portions **51** each include a lock hole **55** that passes the second lock portion **51**. When the first lock portions **47** are fitted into the lock holes **55**, the second lock portions **51** are locked to the first lock portions **47** of the first outer conductor **41**.
(69) As shown in FIG. **3**, the protruding portions **52** project upward from the upper face of the outer conductor body **50**. The protruding portions **52** are provided at the rear end portion of the outer conductor body **50**.
(70) As shown in FIG. **3**, the front stop portion **53** is arranged forward of the outer conductor body **50**. The front stop portion **53** restricts forward movement of the second dielectric **62** arranged inside the second outer conductor **42**.
(71) As shown in FIG. **3**, the partner connection portions **54** are supported by the outer conductor body **50**. Portions of the outer conductor body **50** are cut out. The partner connection portions **54** are arranged in these cutout portions. The partner connection portions **54** are cantilevered due to the rear end portions thereof being supported by peripheral edge portions of the cutout portions of the outer conductor body **50**. The partner connection portions **54** can undergo bending deformation. The partner connection portions **54** elastically come into contact with the partner outer conductor **93** (see FIG. **5**) of the partner connector **90** so as to be electrically connected thereto. The partner connection portions **54** each have a guiding face **56** that guides the partner outer conductor **93** during connection to the partner outer conductor **93**. The guiding faces **56** are formed at the front end portions of the partner connection portions **54**. The guiding faces **56** are inclined radially outward while extending rearward. The guiding faces **56** guide the partner outer conductor **93** outward in the radial direction of the partner connection portions **54**. The radially outward sides of the partner connection portions **54** are electrically connected to the partner outer conductor **93**.
(72) Configuration of Second Dielectric **62**
(73) As shown in FIG. **6**, the second dielectric **62** is arranged between the second inner conductor **22** and the second outer conductor **42**. The second dielectric **62** is shaped as a tube (more specifically, a cylinder).
(74) Other Configurations
(75) The sleeve **70** shown in FIG. **5** is shaped as a tube (more specifically, a cylinder). The sleeve **70** is made of a metal, for example. The first seal member **71** and the second seal member **72** shown in FIG. **5** are shaped as a tube (more specifically, a cylinder). The first seal member **71** and the second seal member **72** are made of rubber, for example. The first seal member **71** is attached to the outer surface of the electric wire **80**. The second seal member **72** is attached to the outer surface of the inner hood portion **15** of the housing **11**. The retaining member **73** is a member for preventing the first seal member **71** arranged in the housing **11** from coming off. As shown in FIGS. **1** and **5**, the retaining member **73** includes an insertion hole **74** and second retaining and locking portions **75**. The electric wire **80** is inserted into the insertion hole **74**. The second retaining and locking portions **75** are locked to the first retaining and locking portions **18** of the housing **11**.
(76) Assembly of Connector **10**
(77) The following description is given mainly with reference to FIG. **5**. First, the retaining member **73**, the first seal member **71**, and the sleeve **70** are attached to the electric wire **80** in this order from the leading end side. The sheath **84** is then removed from the leading end portion of the electric wire **80** so as to expose the shield layer **83**. The insulator **82** is then removed from a portion of the electric wire **80** further closer to the leading end so as to expose the inner conductor **81**. The exposed inner conductor **81** is crimped in the barrel portion **26** of the first inner conductor **21**.

(78) The first inner conductor **21** is inserted into the cavity **63** of the first dielectric **61** from below. The first inner conductor **21** is inserted into the cavity **63** in such a manner that the stabilizer **25** is fitted into the stabilizer fitting groove **67** of the first dielectric **61**. In the process of inserting the first inner conductor **21** into the cavity **63**, the locking portion **24** of the first inner conductor **21** is fitted into the guide groove **66** formed in the inner peripheral surface of the cavity **63**, and slides upward along the guide groove **66**. When fitted in the guide groove **66**, the locking portion **24** undergoes bending deformation due to receiving reaction force from the bottom face of the guide groove **66**. When the first inner conductor **21** has been inserted to the normal insertion position, the locking portion **24** enters the locking hole **64** in communication with the guide groove **66** due to elastic return force. As a result, the first inner conductor **21** is locked to the first dielectric **61** and is restricted from coming downward out of the cavity **63**. In the state where the locking portion **24** has entered the locking hole **64**, the opening between the pair of first connection portions **31** faces the entry hole **65** of the first dielectric **61**.
(79) The first dielectric **61** is inserted into the first outer conductor **41** from below. When the first dielectric **61** has been inserted to the normal insertion position, the entry hole **65** is aligned with the conductor-side fitting hole **45** of the first outer conductor **41** in the front-rear direction. The outer peripheral surface of the tubular portion **44** of the first outer conductor **41** is covered by the exposed shield layer **83** and crimped by the sleeve **70**. As a result, the first outer conductor **41** is electrically connected to the shield layer **83** of the electric wire **80**.
(80) The first outer conductor **41** is inserted into the housing body **12** of the housing **11** from below. When the first outer conductor **41** has been inserted to the normal insertion position, the conductor-side fitting hole **45** of the first outer conductor **41** is aligned with the fitting hole **13** of the housing **11** in the front-rear direction, and the conductor-side fitting groove **46** of the first outer conductor **41** is aligned with the fitting groove **14** of the housing **11** in the front-rear direction. As shown in FIG. **9**, the first lock portions **47** are arranged so as to face the rear side of the fitting hole **13**. After insertion of the first outer conductor **41**, the second retaining and locking portions **75** of the retaining member **73** are locked to the first retaining and locking portions **18** of the housing **11**.
(81) The second inner conductor **22** is inserted into the second dielectric **62** from behind. When the second inner conductor **22** has been inserted to the normal insertion position, movement of the second inner conductor **22** in the front-rear direction relative to the second dielectric **62** is restricted by the inner-conductor-side protruding portions **37** and the retaining projections **38**. The second dielectric **62** is inserted into the second outer conductor **42** from behind. The second dielectric **62** restricted from moving forward upon abutting against the front stop portion **53** of the second outer conductor **42**. The second outer conductor **42** is fitted into the fitting hole **13** of the housing **11** from the front side in such a manner that the protruding portions **52** fit into the fitting groove **14** of the housing **11**. As the fitting of the second outer conductor **42** progresses, the second outer conductor **42** is fitted into the conductor-side fitting hole **45** of the first outer conductor **41**, and the protruding portions **52** of the second outer conductor **42** are fitted into the conductor-side fitting groove **46** of the first outer conductor **41**.
(82) In the process in which the second outer conductor **42** is fitted into the conductor-side fitting hole **45** in the accommodation portion **43**, the second lock portions **51** are pressed by the first lock portions **47** so as to undergo bending deformation in the inward direction. As the fitting progresses further, the first lock portions **47** are fitted into the lock holes **55** of the second lock portion **51**, and the second lock portions **51** return to their original shape due to elastic return force. As a result, the second lock portions **51** are locked to the first lock portions **47**.
(83) When the second lock portions **51** are locked to the first lock portions **47**, the second outer conductor **42** is coupled to the first outer conductor **41**. The first outer conductor **41** is configured so as not to come out from the inside of the housing body **12** to the fitting hole **13**. For this reason, even if the second outer conductor **42** that is coupled to the first outer conductor **41** is pulled in the direction of coming out of the fitting hole **13**, the first outer conductor **41** is caught in the housing

body **12**. In other words, in the state where the second lock portions **51** are locked to the first lock portions **47**, the second outer conductor **42** is retained in the fitting hole **13**.
(84) In the process in which the second outer conductor **42** is fitted into the conductor-side fitting hole **45** in the accommodation portion **43**, the second connection portion **36** of the second inner conductor **22** enters the entry hole **65** of the first dielectric **61**, and moves forward while spreading apart the pair of first connection portions **31**. In the state where the second inner conductor **22** has been normally connected to the first inner conductor **21**, the second inner conductor **22** is sandwiched between the pair of first connection portions **31** of the first inner conductor **21**, and the leading end of the second connection portion **36** of the second inner conductor **22** is arranged inward of the receding face **24**B. At this time, the leading end of the second connection portion **36** of the second inner conductor **22** does not come into contact with the receding face **24**B.
(85) Note that a later-described second outer conductor **42**B, which is different from the second outer conductor **42**, can be fitted into the accommodation portion **43** of the first outer conductor **41**. In other words, the connector **10** is configured such that any one of a plurality of types of second outer conductors (in the present embodiment, either the second outer conductor **42** or the second outer conductor **42**B) can be coupled to the first outer conductor **41**. The second outer conductor **42**B includes an outer conductor body **50**B, second lock portions **51**B, protruding portions **52**B, and partner connection portions **54**B.
(86) The outer conductor body **50**B is shaped as a tube (more specifically, a cylinder) that extends in the front-rear direction. The second lock portions **51**B are arranged rearward of the outer conductor body **50**B.
(87) The second lock portions **51**B are respectively provided on the left and right sides of the second outer conductor **42**B. A second lock hole **55**B is formed in each of the second lock portions **51**B. The second lock portions **51**B have the same shape as the second lock portions **51**.
(88) The protruding portions **52**B are provided on the outer peripheral surface of the outer conductor body **50**B. The protruding portions **52**B project upward from the upper end portion of the outer peripheral surface of the outer conductor body **50**B. The protruding portions **52**B have the same shape as the protruding portions **52**.
(89) The partner connection portions **54**B are supported in a cantilevered manner by the front end of the outer conductor body **50**B, and are shaped so as to project forward. A plurality of (six in the present embodiment) partner connection portions **54**B are provided at equal intervals in the circumferential direction. The partner connection portions **54**B can undergo bending deformation. The partner connection portions **54**B elastically come into contact with the partner outer conductor **93** (see FIG. **5**) of the partner connector **90** so as to be electrically connected thereto. The partner connection portions **54**B each have a guiding face **56**B that guides the partner outer conductor **93** during connection to the partner outer conductor **93**.
(90) The guiding faces **56**B are formed at the front end portions of the partner connection portions **54**B. The guiding faces **56**B are inclined radially inward while extending rearward. The guiding faces **56**B guide the partner outer conductor **93** inward in the radial direction of the partner connection portions **54**B. The radially inward sides of the partner connection portions **54**B are electrically connected to the partner outer conductor **93**.
(91) In other words, the second lock portions **51** and the second lock portions **51**B have the same shape as each other, and are shaped so as to be locked to the first lock portions **47**. For this reason, when either the second outer conductor **42** or the second outer conductor **42**B is selected and fitted into the accommodation portion **43** of the first outer conductor **41**, the second lock portions of the fitted second outer conductor are locked to the first lock portions **47** of the first outer conductor **41**, and the second outer conductor is coupled to the first outer conductor **41**. As a result, the second outer conductor is electrically connected to the first outer conductor **41**. On the other hand, the partner connection portions **54** and the partner connection portions **54**B have different shapes from each other, and are shaped so as to be connected to partner connection portions that have different

shapes from each other. For this reason, the partner connector that corresponds to the second outer conductor that is coupled to the first outer conductor **41** can be fitted to the connector **10**.

(92) Effects of Connector **10**

(93) The first outer conductor **41** of the connector **10** is a member formed with a tubular shape by casting or cutting, and therefore the first outer conductor **41** can be formed so as to suppress the formation of a gap, thus making it possible to improve the shielding performance of the first outer conductor **41**.

(94) Here, if the second outer conductor **42** is also a member formed with a tubular shape by casting or cutting, both the first outer conductor **41** and the second outer conductor **42** do not easily deform, and thus can conceivably be coupled to each other by press fitting. However, in the case of coupling by press fitting, the dimensional tolerance between the first outer conductor **41** and the second outer conductor **42** needs to be reduced in order to ensure electrical connection reliability, which may increase difficulty in manufacturing the outer conductor **40**. In view of this, according to the connector **10**, the second outer conductor **42** is a plate-shaped member and includes the second lock portions **51** that can undergo bending deformation. Therefore, if the second lock portions **51** undergo bending deformation in order to become locked to the first lock portions **47** of the first outer conductor **41**, the dimensional tolerance between the first outer conductor **41** and the second outer conductor **42** can be large, the outer conductor **40** can be manufactured easily, and it is possible to realize coupling with high electrical connection reliability. Also, due to being plate-shaped, the second outer conductor **42** can be manufactured at low cost.

(95) Moreover, a portion of the second outer conductor **42** enters the accommodation portion **43**, thus making it possible to reduce the height of the connector **10** in the direction in which the second outer conductor **42** projects from the first outer conductor **41**.

(96) Also, the second lock portions **51** of the second outer conductor **42** are arranged at positions that close the through holes **48** when the second outer conductor **42** is fitted to the accommodation portion **43**. This therefore makes it possible to suppress a reduction in the shielding performance of the outer conductor **40**.

(97) Also, the second outer conductor **42** includes the partner connection portions **54** that come into contact with the partner outer conductor of the partner connector **90**. For this reason, according to the connector **10**, it is possible to improve the electrical connection reliability between the second outer conductor **42** and the partner outer conductor **93**. Also, due to the second outer conductor **42** being a plate-shaped member, it is possible to easily form the partner connection portions **54** that are capable of elastic contact.

(98) Also, the connector **10** has a configuration in which either one of the two types of second outer conductor **42** and **42**B can be selected and coupled to the first outer conductor **41**, and the two types of second outer conductors **42** and **42**B respectively include the partner connection portions **54** and **54**B that have different shapes from each other. This therefore makes it possible to manufacture a plurality of types of connectors that can be fitted to various types of partner outer conductors while also including the same first outer conductor **41**.

(99) Also, in the connector **10**, the locking portion **24** is locked to the locking hole **64**, thus suppressing the case where the first inner conductor **21** comes out of from the first dielectric **61**. Moreover, the locking portion **24** is shaped as a double-supported beam whose upper and lower end portions are supported by the first inner conductor body **23**. For this reason, according to this connector **10**, a decrease in the impedance of the first inner conductor **21** can be suppressed more than in the case of a configuration in which the locking portion is a lance.

(100) Also, the locking portion **24** is curved. For this reason, according to the connector **10**, it is possible to reduce the insertion force required when inserting the first inner conductor **21** into the first dielectric **61**.

(101) Also, the locking hole **64** and the entry hole **65** of the first dielectric **61** are arranged coaxially with each other via the cavity **63**. For this reason, the locking hole **64** and the entry hole **65** can be

punched out at the same time using the same straight die during manufacturing of the connector **10**.
(102) Also, in a state where the second inner conductor **22** is normally connected to the first inner conductor **21**, the leading end of the second connection portion **36** of the second inner conductor **22** is arranged inward of the receding face **24**B. For this reason, if the first inner conductor **21** is in a partially-inserted state, the leading end of the second inner conductor **22** is abutted against the first inner conductor **21**. Therefore, it can be easily determined whether or not the first inner conductor **21** is in the partially-inserted state. In particular, in the present embodiment, in the state where the leading end of the second inner conductor **22** is abutted against the first inner conductor **21**, the second lock portions **51** of the second outer conductor **42** do not become locked to the first lock portions **47** of the first outer conductor **41**. For this reason, based on this non-locked state, it is possible to more easily determine that the first inner conductor **21** is in the partially-inserted state.
(103) Also, the guide groove **66** is formed on the inner peripheral surface of the cavity **63**, and the guide groove **66** extends along the up-down direction and is in communication with the locking hole **64**. For this reason, according to the connector **10**, while being inserted into the cavity **63**, the locking portion **24** of the first inner conductor **21** can be guided to the locking hole **64** by the guide groove **66**.
(104) Also, in the state where the first outer conductor **41** is located at the normal insertion position, the first lock portions **47** of the connector **10** are arranged so as to face the back side in the fitting direction of the second outer conductor **42** in the fitting hole **13**. The second lock portions **51** then become locked to the first lock portions **47**. In the state where the second lock portions **51** are locked to the first lock portions **47**, the second outer conductor **42** is retained in the fitting hole **13**. In opposite terms, according to this connector **10**, if the first outer conductor **41** is in the partially-inserted state, the positions of the second lock portions **51** and the first lock portions **47** are not aligned with each other, and thus the second lock portions **51** do not become locked to the first lock portions **47**, and the second outer conductor can come out of the fitting hole. For this reason, according to this connector **10**, it is possible to suppress the case where the first outer conductor **41** and the second outer conductor **42** become locked to each other while in the partially-inserted state.
(105) Also, the housing **11** includes the fitting groove **14** that is formed in the inner peripheral surface of the fitting hole **13** and extends along the fitting direction of the second outer conductor **42**, and the second outer conductor **42** includes the protruding portions **52** that fit into the fitting groove **14** in the process of fitting to the first outer conductor **41**. For this reason, according to the connector **10**, the second outer conductor **42** can be positioned relative to the housing **11** in the circumferential direction.
(106) Also, the housing **11** is L-shaped and does not including a housing lock portion by which the first outer conductor **41** and the second outer conductor **42** are locked to the housing **11**, and therefore the first outer conductor **41** and the second outer conductor **42** can easily separate from the housing **11** when not coupled to each other. Therefore, according to this configuration, it is easy to check whether or not the first outer conductor **41** and the second outer conductor **42** are correctly coupled.
(107) Also, in the connector **10**, in the state where the second lock portions **51** are locked to the first lock portions **47**, the protruding portions **52** of the second outer conductor **42** are fitted into the conductor-side fitting groove **46** of the first outer conductor **41**, and therefore the second outer conductor **42** can be positioned relative to the first outer conductor **41** in the circumferential direction.
(108) Also, the shield layer **83** of the electric wire **80** is electrically connected to the first outer conductor **41**. The first outer conductor **41** is electrically connected to the second outer conductor **42**, and the partner outer conductor **93** of the partner connector **90** is electrically connected to the second outer conductor **42**. The second outer conductor **42** extends in a direction intersecting (more specifically, orthogonal to) the extending direction of the first outer conductor **41**. For this reason, according to this connector **10**, it is possible to change the route in the direction intersecting the

extending direction of the electric wire **80**.
Other Embodiments of Present Disclosure
(109) The embodiments disclosed in the present embodiment are illustrative in all respects and not intended to be construed as limiting.
(110) (1) Although the connector is L-shaped in the above embodiment, the connector does not need to be L-shaped. For example, the connector may be I-shaped (have a straight shape).
(111) (2) Although the second outer conductor closes the through hole of the first outer conductor in the above embodiment, a configuration is possible in which the second outer conductor does not close the through hole.
(112) (3) Although the inner conductor is constituted by a plurality of (specifically, two) members (the first inner conductor and the second inner conductor) in the above embodiment, the inner conductor may be constituted by one member.
(113) (4) Although the partner connection portion is configured to come into elastic contact with the partner outer conductor in the above embodiment, the partner connection portion does not need to come into elastic contact with the partner outer conductor.
(114) (5) Although the first inner conductor includes a locking portion in the above embodiment, a configuration is possible in which the first inner conductor does not include a locking portion. Also, the locking portion does not need to be curved.
(115) (6) Although the entry hole is arranged coaxially with the locking hole in the above embodiment, the entry hole and the locking hole do not need to be arranged coaxially.
(116) (7) Although the leading end of the second inner conductor is arranged inward of the receding face of the locking portion in the first inner conductor in the above embodiment, the leading end of the second inner conductor does not need to be arranged inward of the receding face. For example, the leading end of the second inner conductor may be arranged outward of (forward of) the open end of the receding face.
(117) (8) Although the leading end of the second inner conductor does not come into contact with the receding face of the locking portion of the first inner conductor in the above embodiment, the leading end of the second inner conductor may come into contact with the receding face.
(118) (9) Although the guide groove is formed on the inner peripheral surface of the cavity in the above embodiment, a configuration is possible in which the guide groove is not formed.
(119) (10) Although the electric wire is a coaxial cable in the above embodiment, the electric wire does not need to be a coaxial cable, and may be a cable for differential signal transmission, for example.
(120) (11) Although only a portion of the second outer conductor is inserted into the accommodation portion of the first outer conductor in the above embodiment, the entirety of the second outer conductor may be inserted into the accommodation portion.
(121) From the foregoing, it will be appreciated that various exemplary embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various exemplary embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A connector comprising: an inner conductor; an outer conductor configured to surround the inner conductor; and a housing made of an insulator and configured to surround the outer conductor, wherein the outer conductor includes: a first outer conductor configured to be electrically connected to a shield layer of an electric wire, and a second outer conductor configured to be electrically connected to the first outer conductor and electrically connected to a partner outer conductor of a partner connector, the first outer conductor is a member formed with a tubular shape

by casting or cutting, and includes a first lock portion and an accommodation portion open in one end side of the first outer conductor, the second outer conductor is a plate-shaped member and includes a second lock portion configured to be locked to the first lock portion, and a portion of the second outer conductor enters the accommodation portion.

2. The connector according to claim 1, wherein one of a plurality of types of the second outer conductor is selectively coupled to the first outer conductor, and the plurality of types of second outer conductors include partner connection portions having different shapes from each other.

3. The connector according to claim 1, wherein the first lock portion projects inward from an inner peripheral face of the accommodation portion, and the second lock portion projects rearward from the second outer conductor and includes a lock hole that passes through the second lock portion.

4. A connector comprising: an inner conductor; an outer conductor configured to surround the inner conductor; and a housing made of an insulator and configured to surround the outer conductor, wherein the outer conductor includes: a first outer conductor configured to be electrically connected to a shield layer of an electric wire, and a second outer conductor configured to be electrically connected to the first outer conductor and electrically connected to a partner outer conductor of a partner connector, the first outer conductor is a member formed with a tubular shape by casting or cutting, and includes a first lock portion, and the second outer conductor is a plate-shaped member and includes a second lock portion configured to be locked to the first lock portion and a plurality of partner connection portions electrically connected to the partner outer conductor of the partner connector and cantilevered due to rear end portions of the plurality of partner connection portions being supported by peripheral edge portions of cutout portions of the second outer conductor.